



**DEPARTMENT OF THE AIR FORCE
AIR FORCE LIFE CYCLE MANAGEMENT CENTER
WRIGHT-PATTERSON AIR FORCE BASE OHIO**

MEMORANDUM FOR RECORD

FROM: AFLCMC/ROU
Human Systems Program Office
1981 Monahan Way
Wright-Patterson AFB, OH 45433

SUBJECT: AFLCMC/ROU Safe-to-Fly (StF) Test Plan for Commercial and Government Off-The-Shelf (COTS, GOTS) Gloves

1. The purpose of this Test Plan is the certification of gloves worn by Air Force (AF) aircrew as Safe-to-Fly (StF). Any application of the below test methods to articles, not readily identified as handwear for AF use, is in violation of its designated use. If applied to articles not readily identified as footwear, all risks and consequences associated with its use in the environment become the responsibility of the individual user(s). All test data and results provided to support COTS StF determinations shall be within one year from the date of submission.

2. STF Definition. All AF StF handwear must conform, at minimum, to the following:

Note: NSN gloves managed by Defense Logistics Agency (DLA), including those for other services, can be designated AF StF after a review of the technical requirements in the associated Purchase Description (PD) or Military Specification (MILSPEC) and a StF determination made by the Aircrew Performance Material Leader, equivalent, or appropriately delegated authority.

Note: The primary purpose of the gloves is to provide protection against flames and other hand injuries while allowing the aircrew member to accomplish assigned duties within and around the aircraft, even in adverse conditions.

a. Color. Glove colors are primarily Sage Green (1641), Foliage Green (504), Desert Tan (499), or Coyote (498). Other colors or combinations of colors, different cloth shell and leather colors such as green fabric with black leather, may be allowed by specification, PD or specific approval of the Aircrew Performance Material Leader, equivalent or appropriately delegated authority.

b. Touchscreen. Touchscreen capable leather is required on AF StF gloves. Members shall be able to operate Electronic Flight Bag devices, iPads, smartphones, and other capacitive touchscreen devices, while wearing these gloves.

c. Flame Retardant (FR). All AF StF gloves are required to be Flame Resistant in

accordance with ASTM D6413 (Test No. 191A, Method 5903). This includes the TS leather and all exterior components of the glove. Underlayers and next-to-skin layers must self-extinguish and not melt nor drip. A modified version of the vertical flame test, where the finished glove material is folded and each end of the specimen is held into the vertical flame chamber in the specimen holder should be completed. The sample naturally forms a “u” shape in the specimen holder. The lower part of the fold of the “u” is placed at the exact same height above the flame as the standard vertical sample placement as required in the test method.

d. Water Resistance. The gloves shall be tested for their resistance to rain or water spray in accordance with AATCC 127 Water Resistance: Hydrostatic Pressure Testing.

e. Colorfastness. The gloves shall be tested for colorfastness with AATCC-8 (Colorfastness to Crocking), AATCC-62 (Colorfastness to Laundering) and ASTM-D-6012 (Determination of Resistance of Color Leather to Bleeding).

3. Type Glove temperature ranges and associated specifications (MILSPEC and PD).

a. Hot weather (>70°F): Hot weather StF gloves shall comply with the requirements listed in FQ PD-09-02, Gloves, Flyer’s Summer, Type GS/FRP-2 or MIL-DTL-81188. Hot weather gloves are lightweight, FR, unlined (or minimally lined), water resistant, highly dexterous gloves with excellent manipulation of small objects and a good grip. They are often used in Temperate weather conditions.

b. Temperate Weather (≥ 32°F to 70°F): Temperate Weather StF gloves comply with some of the requirements listed in PD FQ PD-09-02, Gloves, Flyer’s Summer, Type GS/FRP-2 and/or MIL-DTL-32173A, Glove, Flyer’s, Intermediate Cold, HAU-15/P. Temperate weather gloves are FR and usually are thicker (heavier) than a Type GS/FRP-2 glove. They may have a lining, with or without a small amount of insulation. They may have reinforcing (padded or unpadded) leather at vulnerable sections of the hand (heel of palm, knuckles, etc.). They may be water and/or POL resistant and/or proof. While generally not as dexterous as the Type GS/FRP-2 glove, temperate weather gloves have good dexterity and manipulation of small objects, a good grip and are often subjected to rough use by the user.

c. Intermediate Cold/Wet Weather (14°F to 32°F): Intermediate cold weather StF gloves shall comply with the requirements of MIL-DTL-32173A, Glove, Flyer’s, Intermediate Cold, HAU-15/P StF gloves. They are TS, FR, water resistant gloves, with some dexterity and ability to manipulate small objects and good grip. They may also be subjected to some rough use.

d. Cold weather (-20° to 32°F): Cold weather StF gloves shall comply with the requirements of MIL-DTL-32094A, Glove, Flyer’s, Cold Protective, HAU-14/P. These are TS, FR, Waterproof gloves that typically have little dexterity or manipulation capability, but some grip.

e. Extreme Cold weather (-20°F to -49°F, or below): Extreme cold weather gloves shall comply with the requirements of MIL-DTL-32094A, Glove, Flyer's, Cold Protective, HAU-14/P. These are TS, FR, Water-resistant gloves with enhanced cold protection for the temperature ranges noted.

f. Glove Insulation: The amount and type of insulation used will be stated by the vendor/manufacturer. The "clo" (average insulation) value (ASTM C518 Standard Test Method for Steady State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus) for the insulation used will be determined and reported by the vendor/manufacturer.

g. The whole glove clo, tested on a thermal instrumented hand model in accordance with ASTM F1291 Standard Test Method for Measuring the Thermal Insulation of Clothing Using a Heated Manikin, will be tested by the manufacturer, or possibly by the USAF depending on annual test schedule timing and pending availability and authorization of funding. A baseline standard glove of which the proposed glove is intended to replace or improve upon may also be tested concurrently with the proposed candidate.

h. Certificate of Compliance (C of C) from a manufacturer's vendor certifying whole-glove, components, and/or materials meet a PD or MILSPEC requirement, in lieu of a specific test report, may be submitted for consideration for acceptance depending on the intended use and established requirements for each glove type. Data supporting the C of C may be requested.

4. The manufacturer shall identify whether the glove model/style submitted for certification meets Berry Amendment (10 U.S.C 2533a) criteria. Only Berry Amendment compliant materials/capabilities or Berry waived materials may be submitted for consideration for acceptance.

5. After the initial, or subsequent three-year StF approval, and at the discretion of the Senior Material Leader representative, vendors may submit a recertification memo stating the glove make/model has not changed in materials, nor processing practices, and is still fully Berry Amendment compliant. Reasons for requiring a full StF submission, with sample gloves and all required testing data, versus a memo, are noted discrepancies from the field or other noted or reported discrepancies with the materials and/or processing practices for the glove.

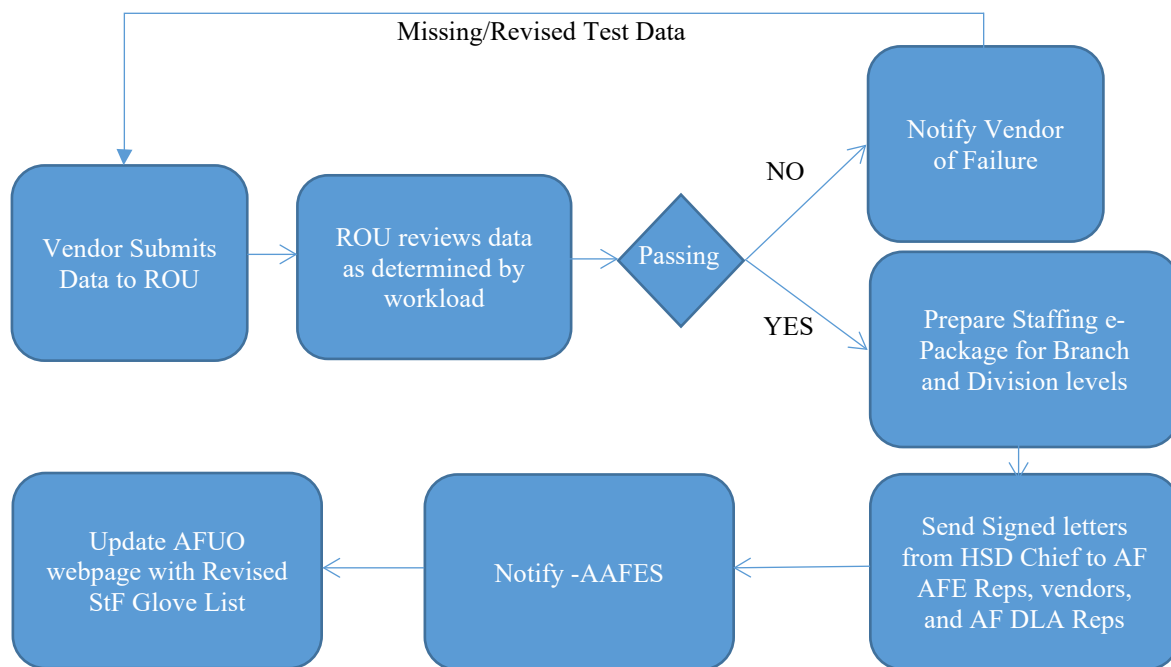
6. The Aircrew Performance Material Lead Engineer POC for this StF plan is Mr. Don Fleury at (508) 206-3071 or donald.f.fleury.civ@army.mil.

NICHOLE L. TAYLOR, NH-04, DAF
Chief Engineer, Human Systems Division
AFPEO Combat Readiness

Attachment:
ROU Process for StF Gloves

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ROU Process for StF Gloves

The ROU process for StF COTS gloves is outlined in the graphic below. Vendors provide data and glove samples at their convenience. The data (not older than 1-year from submission date) is reviewed and compared to the StF requirements by the assigned engineer(s). A package is prepared that establishes the gloves meet all required StF criteria as outlined in the StF plan and coordinated for review and approval. The package includes a glove StF checklist (detailing the requirements and referencing the testing results), testing/pass/fail results, a MAJCOM StF glove memo, a Vendor StF glove memo and a revised AF StF Glove List memo.



A vendor STF letter (cc AAFES) is prepared for each specific manufacturer model/style for a three-year period. Multiple glove model/styles may be included in a single memo. The revised StF list is attached to the MAJCOM letter and uploaded on various Aircrew Flight Equipment (AFE) sites. The letters are signed by the Chief of Human System Division or designated representative. Once the package is approved, signed/dated letters are sent to the manufacturer indicating the specific commercial glove style(s) is approved as StF for three years. Note the vendor letter includes a statement to the effect that any material or manufacturing changes will void the StF approval unless the Vendor coordinates the planned change and obtains AF approval from AF AFLCMC/ROU Materiel Leader and certification package resubmission, beforehand. The MAJCOM StF letter is provided to inform the major AF commands that a new glove has been added to the StF lists or an existing glove has been recertified for an additional three years.

NOTE: New sample gloves, supplanted by the recertification process with newer gloves, or withdrawn by the vendor for consideration, may be returned to the vendor per their request. Otherwise, they will be disposed of.